

CHRIST (Deemed to be University), Bangalore

Centre for Research Projects SEED MONEY SCHEME Project Proposal 2024 – 25

Reference Number (Office use only)	
*Seed Money Scheme Level	01/02/03
*Project title	Analysis of Social Psychology using Signed Graphs
*Duration (in months)	24 months
*Seed Money Requested	1 lakh
*Name, designation & Department, School / Deanery, Campus of the Principal Investigator	James Joseph Assistant Professor Department of Mathematics School of Sciences
Contact Number	9947935481
Email ID	james.joseph@christuniversity.in
Number of publications in Scopus/WoS indexed Journals.	5
*Project Team members & their role in this project	None

1. Project title : Analysis of Social Psychology using Signed Graphs
2. Project summary (250 words) : The proposed project aims to explore the application of signed graphs in understanding and analyzing social relationships and psychological dynamics. Signed graphs, where edges are marked with positive or negative signs representing friendship or antagonism, respectively, offer a robust framework for modeling complex social networks. This research will employ signed graph theory to study group dynamics, conflict resolution, and the spread of behaviors within social networks. By integrating psychological theories with mathematical models, we aim to uncover patterns and insights that are not apparent through traditional social psychology methods. The project will involve data collection from social media, surveys, and existing datasets, followed by the construction and analysis of signed graphs. The findings are expected to contribute to the development of more effective interventions in social conflict resolution and enhance our understanding of social cohesion and polarization.

3. International and National Literature Review status (400 words)

Internationally, the application of graph theory in social psychology has gained traction, particularly in studying social networks, influence spread, and group behaviour dynamics. Studies by Wasserman and Faust (1994) laid the groundwork for network analysis in social sciences, and recent advancements by Leskovec et al. (2010) have integrated large-scale social media data. The concept of signed graphs was introduced by Harary (1953), emphasizing their utility in capturing the dual nature of social relationships. Contemporary research by researchers like Doreian and Mrvar (2009) has expanded the use of signed graphs to study social balance and conflict within networks.

Nationally, the adoption of graph theory in social psychology is still emerging. Indian researchers have primarily focused on network analysis in fields like sociology and anthropology. However, there is a growing interest in integrating mathematical models with social science research. Studies by Desai et al. (2018) and Singh et al. (2021) have started exploring network analysis in Indian social contexts, but the specific application of signed graphs remains underexplored.

4. Gap area(s) identified between International and National Literature reviews (200 words)

The primary gap identified is the limited application of signed graphs in social psychology, particularly within the Indian context. While international research has advanced in this domain, national studies have yet to fully leverage signed graphs to explore social and psychological phenomena. There is also a lack of interdisciplinary research combining mathematical models with social psychology theories, creating a gap in comprehensive approaches to understanding social networks.

5. Problems intended to be addressed by proposed project (*establish relation between project objectives and identified gap areas*) (200 words)

The proposed project aims to address the gap by applying signed graph theory to analyse social relationships and psychological dynamics within Indian social contexts. Specifically, the project will focus on understanding group dynamics, conflict resolution, and behaviour spread. By linking the project objectives to the identified gaps, we intend to develop a nuanced understanding of social networks, providing insights that can inform interventions and policies to enhance social cohesion and mitigate conflicts.

6. Objectives of the Proposed Study (200 words)

To construct signed graphs representing social networks from collected data.

To analyse the structure and dynamics of these signed graphs to identify patterns of social cohesion and conflict.

To explore the relationship between signed graph properties and psychological phenomena such as influence, trust, and antagonism.

To develop interventions based on the findings to enhance group dynamics and conflict resolution strategies.

7. Methods proposed for project (300 words)

The project will employ a mixed-methods approach, combining quantitative and qualitative data collection with mathematical modelling.

Data Collection:

- 1) Surveys and Questionnaires: To gather information on social relationships, perceptions, and behaviors from participants.
- 2) Social Media Data Mining: To extract large-scale social network data from platforms like Twitter and Facebook.
- 3) Existing Datasets: To utilize available datasets for comprehensive analysis.

Graph Construction:

- 1) Data Processing: Cleaning and pre-processing data to construct signed graphs.
- 2) Graph Theory Application: Employing algorithms and models to analyze the properties of the constructed signed graphs.

Analysis:

- 1) Statistical Analysis: To identify significant patterns and correlations within the data.
- 2) Graph Metrics: To measure properties such as centrality, balance, and clustering in the signed graphs.

Interventions:

- 1) Developing strategies and tools for conflict resolution and enhancing group dynamics based on the analysis results.

8. Activity / work plan

Objectives	Activities/ Work	Expected outcome	Person/Area responsible	Remarks
Construct signed graphs	Data collection, preprocessing, and graph construction	Signed graphs representing social networks	self	
Analyze graph structure	Application of graph theory and statistical analysis	Identification of social patterns and dynamics	self	
Explore relationships	Correlation of graph properties with psychological phenomena	Insights into social cohesion and conflict	self	
Develop interventions	Formulation of strategies	Effective conflict	self	

	based on findings	resolution tools		
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9. Expected Outputs of the project :

Outputs	Remarks (with details/proposed timeline/publication location)
Number of Stakeholders who will benefit	Researchers, psychologists, policymakers
Number of Scholarly Papers in Journals	2 (Targeted at high-impact journals)
Number of Scholarly Papers in Edited Book/s	0
Publication of Book/s	0
Policy Papers	0
Dataset	Curated datasets for future research
Infrastructure	
Patents	0
Copyrights	0
Modules	0
Models	1 (Graph model for social network analysis)
Prototypes	0
Others ...	-

10. Applicability of proposed project results.

Applicability	Remarks
Usage and future potential of proposed project results	Results will be used to inform social psychology research and interventions
Usage and future potential of proposed project technologies	Technologies for data collection and analysis will be developed
Usage and future potential of proposed project Innovations	Innovative approaches to conflict resolution and group dynamics enhancement
Pathbreaking aspect of Proposed project results	Interdisciplinary integration of graph theory and social psychology
Policy Relevance of the proposed project results	Recommendations for policymakers on social cohesion and conflict mitigation
Others	

11. Potential Risks of the project and Prevention Methods (200 words)

Data Privacy Concerns: Ensuring all data collection adheres to ethical guidelines and participants' privacy is protected.

Data Quality Issues: Implementing rigorous data pre-processing and validation methods to ensure accuracy.

Interdisciplinary Challenges: Facilitating effective communication and collaboration between psychology and mathematics experts.

12. Monitoring and Evaluation Framework (200 words)

Regular Progress Reviews: Monthly meetings to review progress and address any issues.

Milestone Tracking: Establishing key milestones and tracking their achievement.

Feedback Mechanisms: Incorporating feedback from stakeholders and experts to refine approaches.

13. Time line of the project

Month	Activities
1-6	Initial data collection and pre-processing
6-12	Construction of signed graphs
13-18	Analysis of graph structures
18-21	Development of interventions
22-24	Documentation and dissemination of results

14. Impact out of the Project Activities for the society (200 words)

The project aims to provide insights into social dynamics that can enhance community cohesion and mitigate conflicts. By developing effective conflict resolution strategies and tools, the project will contribute to building more harmonious and resilient communities. The findings will inform policies and interventions, ultimately benefiting society by fostering understanding and collaboration.

15. Theory of Change (Please refer the below image and prepare)

Resources:

Materials:

Survey tools and questionnaires

Data mining software

Statistical analysis software

Financial:

Seed Money: INR 2,00,000

Institutional Support:

Access to university libraries and research databases.

Activities

Data Collection:

Conduct surveys and distribute questionnaires.

Mine social media data.

Gather and pre-process existing datasets.

Graph Construction and Analysis:

Build signed graphs from collected data.

Apply graph theory algorithms to analyse the structures.

Correlation and Exploration:

Study the relationship between graph properties and psychological phenomena.

Intervention Development:

Formulate strategies for conflict resolution based on analysis results.

Develop training modules and tools for enhancing group dynamics.

Output

Tangible Evidence:

Constructed signed graphs representing social networks.

Detailed analysis reports on social cohesion and conflict dynamics.

Developed conflict resolution strategies and training modules.

Scholarly papers and publications.

Outcome

Expected Changes:

Improved understanding of social relationships and dynamics through the application of signed graphs.

Enhanced methodologies for social psychology research incorporating mathematical models.

Effective conflict resolution strategies and tools adopted by stakeholders.

Increased collaboration between departments of psychology and mathematics.

Impact

Societal Impact:

Strengthened social cohesion within communities through better conflict resolution.

Informed policies and interventions based on research findings.

Contribution to the academic body of knowledge in social psychology and graph theory.

Enhanced capacity of researchers to utilize interdisciplinary approaches in social sciences.

16. Budget estimates:

Sl No	Items	Quantity	Months/ Years	Cost	Total Cost	Particulars/ Justification	Remarks (Office Use)
A	Recurring						
1	Project Staff						
2	Field Staff						
3	Data Processing			20,000			
4	Others						
	Sub-Total			20,000			
B	Non – Recurring						
1	Equipment & Chemicals						
2	Others			20,000			
	Sub-Total			40,000			
C	Project Activities - Costs (if required)						
1	1						
2	2						
3	3						
4	4						
5	Others						
	Sub-Total						
D	Project Administration Cost						
1	Travel			20,000			
2	Stationery			10,000			
3	Printing			10,000			
4	Others			20,000			
	Sub-Total			60,000			
	Grand Total			1,00,000			
	Total in words: One Lakh Rupees.						

References

- [1] F. Harary, On the notion of balance of a signed graph, Mich. Math. J. 2 (1953), no. 2, 143-146.
- [2] P. Doreian and A. Mrvar, Partitioning Signed Social Networks, Social Networks, 31(2009), no. 1, 1-11.
- [3] J. Leskovec, D. Huttenlocher and J. Kleinberg, Signed Networks in Social Media, CHI '10: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, 2010.
- [4] S. Warreman and K. Faust, Social Network Analysis, Cambridge University Press, 1994.

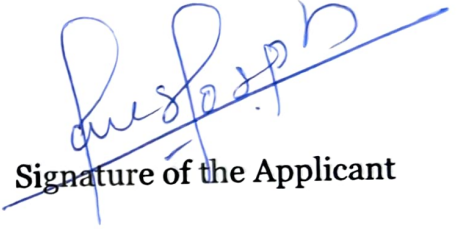
Declaration

I hereby declare that:

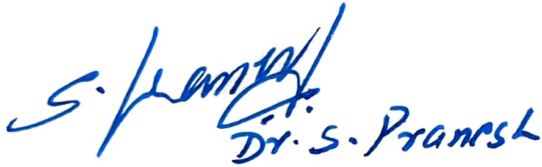
1. I am not a defaulter of any previous Seed Money Scheme of CHRIST (Deemed to be University).
2. I have neither been subjected to any disciplinary action nor found guilty of any offence in my career.
3. The Research/Project Proposal and its contents are entirely original and as per the standard practice.
4. I have not concealed any information in my application. If the University finds any contrary information at any stage, it may cancel the study/project out rightly.

Place: Bengaluru

Date: 05/08/2023



Signature of the Applicant



Dr. S. Prakash

Name & Signature of HoD



Deepthi 5/8/24
Name & Signature of Dean/Director

Associate Dean
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